

Bangladesh University of Business and Technology (BUBT)



LAB Report

Course Code :
Course Title :
Experiment No. :
Experiment Name :

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Intake : 53
Section : 01
Program : B.Sc. in CSE

Name:

Dept. of
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Business and Technology

Date of Submission :

Signature of Teacher

Experiment Name: Static Routing Protocol

Introduction:

Static Routing is a method of manually configuring routes in a router. In this method, the network administrator enters routing information into the routing table manually. Unlike dynamic routing, static routing does not automatically update routes. It is simple, secure, and suitable for small networks. Static routing requires less CPU and bandwidth compared to dynamic routing protocols.

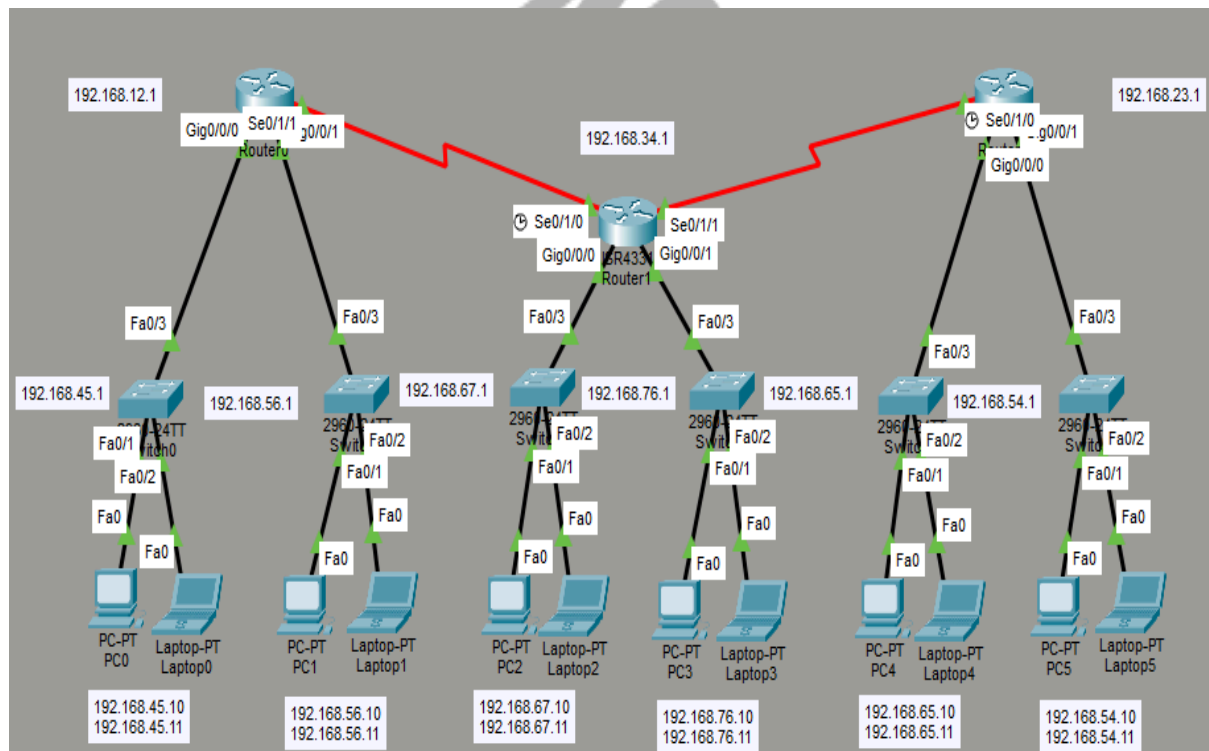
In this experiment, we configured static routes between multiple networks and verified connectivity using ping commands.

Software and Tools Used:

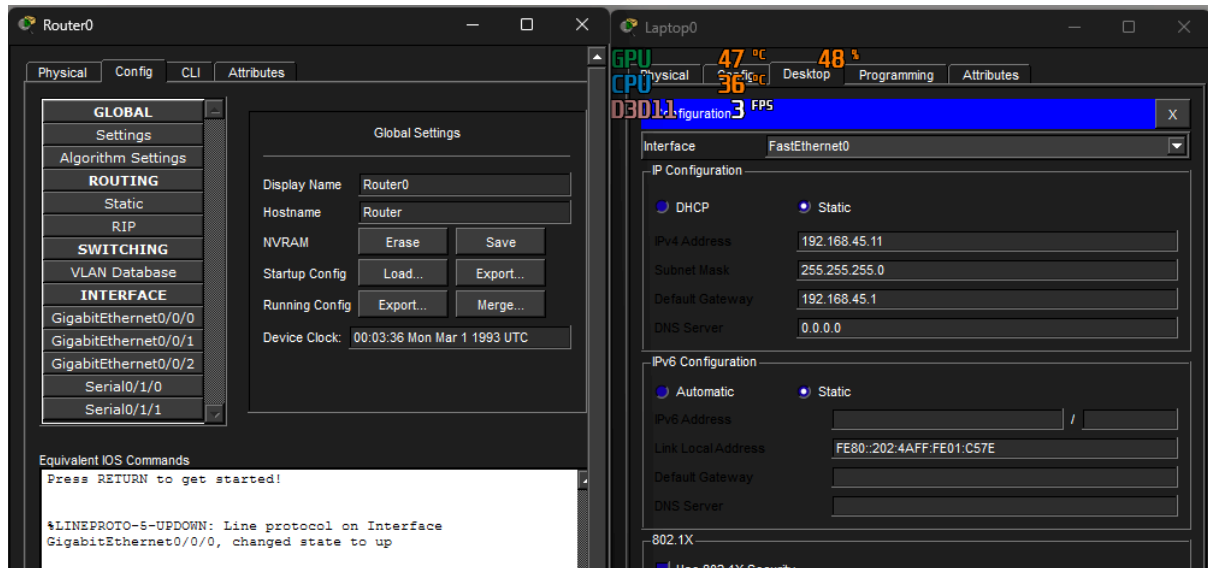
1. Cisco Packet Tracer
2. Switches
3. Routers
4. PCs and Laptops
5. Ethernet Straight-through and Crossover Cables
6. Command Line Interface (CLI)

Procedure:

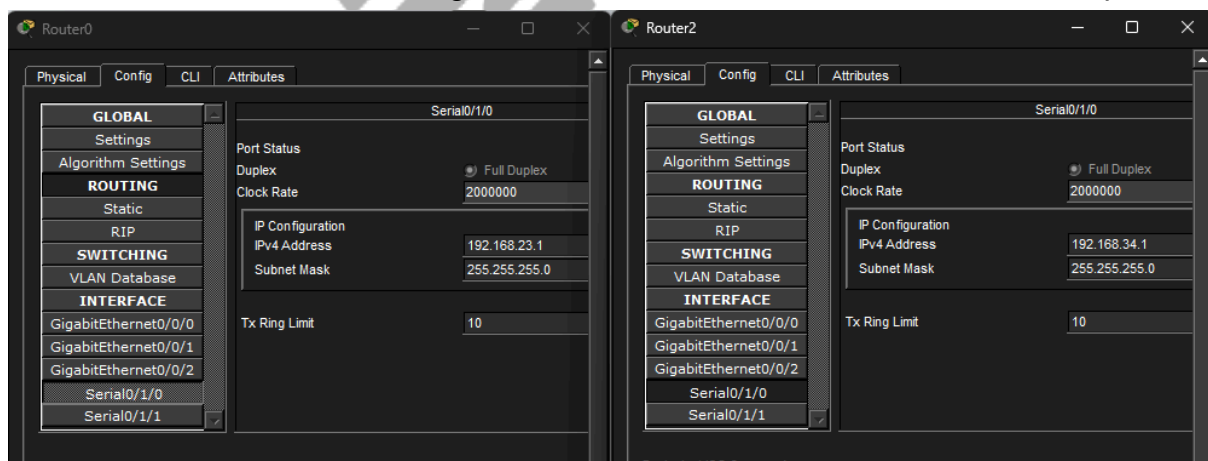
1. Open Cisco Packet Tracer and create the required network connections with routers, switches, and PCs.



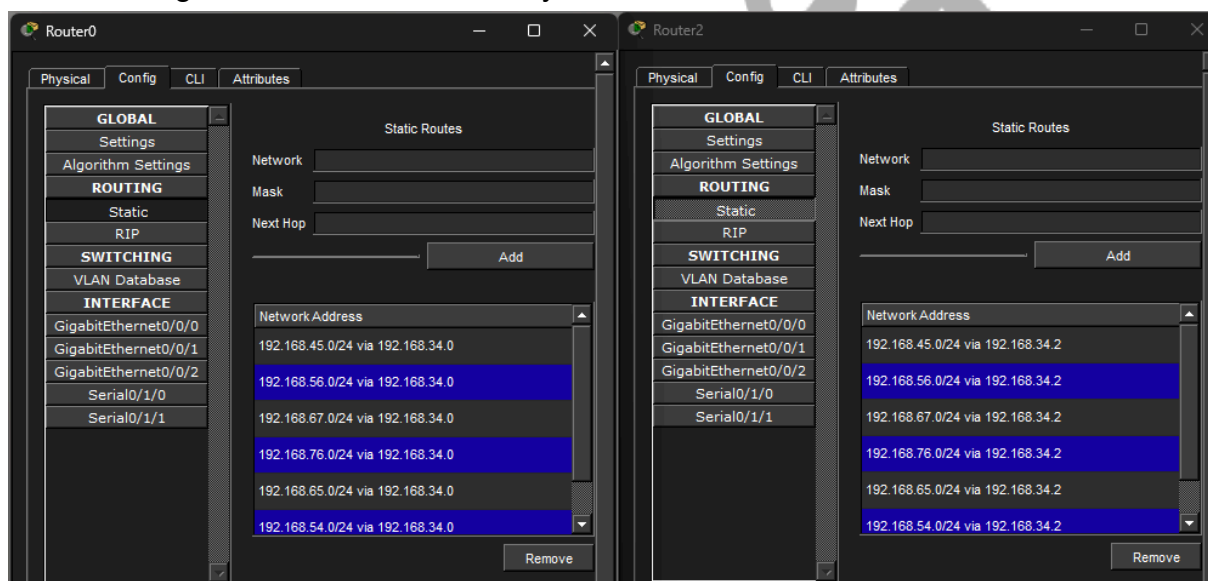
2. Assign appropriate IP addresses to all PCs and router interfaces.



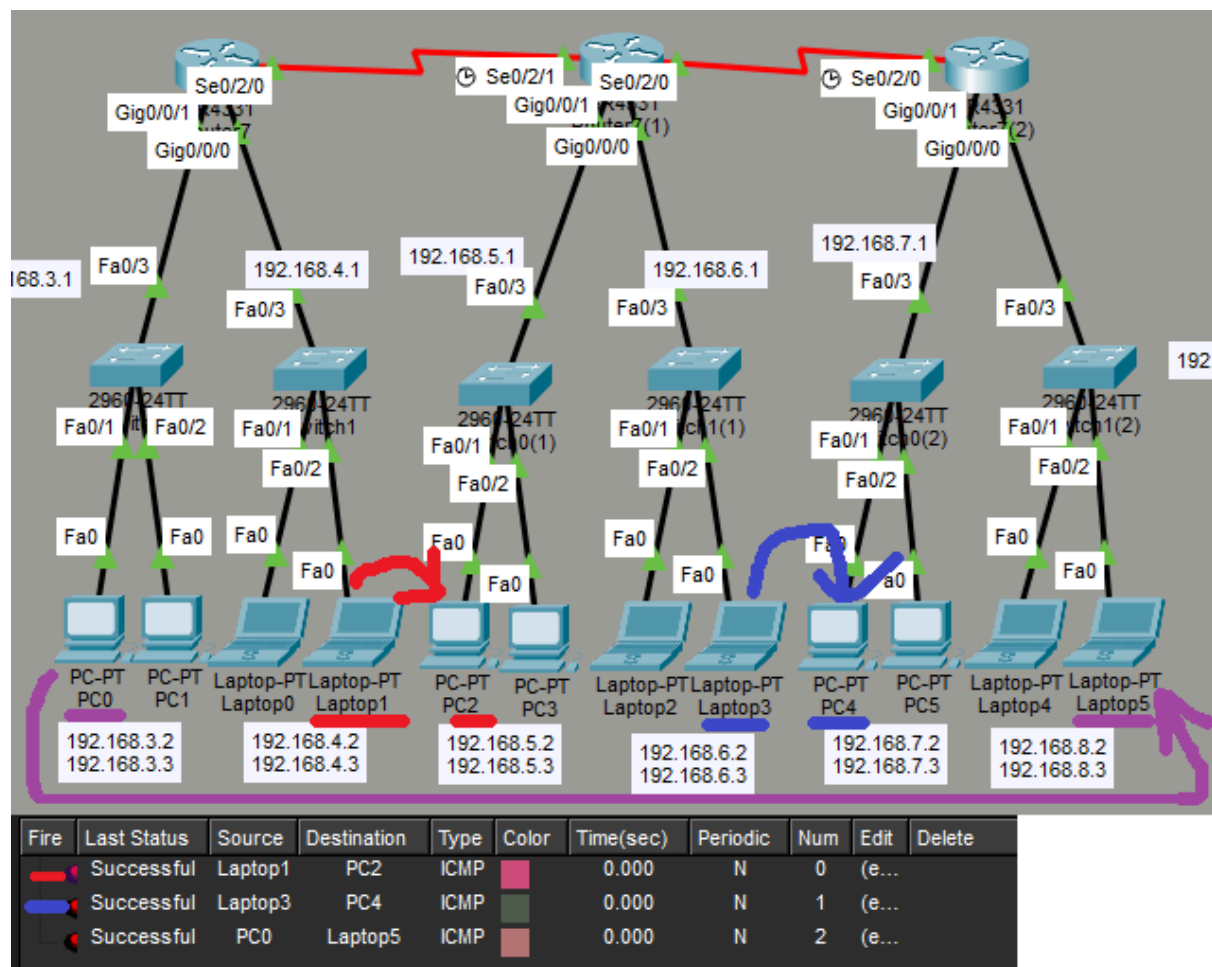
3. Connect the routers using Serial DCE/DTE cables between their serial ports.



4. Configure static routes manually on each router to connect different networks.



- Verify connectivity by sending a ping between PCs of different networks.



Conclusion:

In this experiment, we successfully configured static routing between different networks. We manually added routes in routers and verified connectivity using the ping command.

Static routing is simple and efficient for small networks but not suitable for large networks because it requires manual configuration and maintenance.